

Olist Marketplace Performance & Customer Satisfaction Analysis

Data Analytics Hackathon — Analysis Report

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Prepared for Olist Operations & Customer Experience Leadership

1. Problem Understanding

Olist operates as a marketplace integrator, allowing small and medium Brazilian merchants to sell through major e-commerce channels under a single contract. Sellers fulfill orders through Olist's logistics partners, and customers receive a satisfaction survey (1–5 stars) after estimated or actual delivery.

Leadership holds approximately two years of order history (September 2016 – October 2018) spanning orders, items, payments, products, sellers, customers, reviews, and geolocation — but no single consolidated view connects these dimensions. This analysis was commissioned to answer a direct question: what is actually shaping the customer experience on the platform, and where should Olist focus as it scales to more sellers and more regions of Brazil?

Scope: This report addresses all six core questions defined in the brief — marketplace performance over time, delivery performance and satisfaction, seller and geographic patterns, product category performance, payment behavior, and root cause analysis of low review scores.

2. Analytical Approach

2.1 Data Sources

Nine related CSV files covering roughly 99,441 orders, 112,650 order items, 103,886 payment records, 100,000 review records, 32,951 products, 3,095 sellers, and over 1 million geolocation readings, all released publicly by Olist under a CC BY-NC-SA license.

2.2 Data Cleaning Decisions

- Deduplicated the reviews table to one row per order, keeping the most recently answered review where an order had more than one survey response — the raw table joins many-to-one on order_id and inflates row counts if used directly.
- Resolved multi-payment orders (about 4,400 orders) to a single dominant payment type, selected by the payment record with the highest value, rather than concatenating payment method names into combined categories.
- Aggregated the geolocation table (roughly 52 rows per zip code prefix on average) to one mean latitude/longitude per prefix before joining to customers and sellers, avoiding row multiplication.
- Verified the master table's row count matched the original order count (99,441) after all merges, to catch any unintended row duplication before analysis.
- Restricted delivery-timing analysis to delivered orders, while separately confirming that "delivered" status does not guarantee a complete delivery timestamp, and checking whether that gap was systematically associated with review score.

2.3 Methods by Question

Question	Method
Q1 — Performance over time	Monthly aggregation of order volume, revenue, and average review score
Q2 — Delivery timing & satisfaction	Delivery delay bucketed vs. estimate; cross-cut by state and category, restricted to cells with adequate sample size ($n \geq 15$)
Q3 — Seller & geographic patterns	Review score and volume by state, with a sample-size stability threshold ($n \geq 50$)

Q4 — Category performance	Review score, price, and volume by product category (English-translated labels)
Q5 — Payment behavior	Review score and order value by dominant payment type and installment count
Q6 — Root cause analysis	Random Forest regression on review score, validated on a held-out test set, with a collinearity check between order value and freight

A note on evidence standard: this analysis is observational, not experimental. Findings are reported as associations — factors that move together with review scores — rather than proven causes. Where more than one plausible explanation exists, both are noted.

3. Key Insights

3.1 Q1 — Marketplace Performance Over Time

Monthly order volume grew substantially across the observed period — from a handful of orders in September 2016, through roughly 300–800 orders per month in late 2016 and early 2017, up to a sustained 6,000–6,500 orders per month by mid-2018. Revenue tracked this same growth curve closely, rising from roughly R\$50,000/month in the platform's early months to over R\$850,000/month by mid-2018. Average review score, meanwhile, held in a comparatively narrow band across the period (roughly 4.0–4.3 in the platform's stable months), without a clear downward trend as volume scaled up.

Note on data edges: September 2016, December 2016, and September–October 2018 show very low order counts (4, 1, 16, and 4 orders respectively) and are almost certainly partial or truncated months — likely the platform's launch window and the data extract's cutoff — rather than genuine demand collapses. These edge months were excluded from trend interpretation, and should be excluded or footnoted in any chart shown to leadership.

Do these tell the same story? The correlation between monthly order volume and average review score across the period is **0.472** — a moderate positive relationship. This suggests that as Olist scaled, review scores did not systematically erode; if anything they moved somewhat together with growth, though the relationship is far from perfect (revenue and volume grew roughly 15–20x over the period, while average review score moved by only a few tenths of a point). The practical read: growth has not come at a clear, visible cost to average satisfaction at the platform level — but this masks the more granular findings below, where specific delivery, state, and category patterns show much larger swings than the aggregate trend does.

3.2 Q2 — Delivery Performance and Customer Satisfaction

Delivery delay relative to the estimated delivery date shows a clear, negative relationship with review scores. Importantly, this relationship is broad-based rather than concentrated in a small number of outliers — it holds consistently across the states and product categories with enough order volume to measure reliably.

Sample-size discipline: comparisons required at least 15 delivered, reviewed orders in both the "On Time" and "Very Late" delivery buckets before being reported. Of 27 states, 13 met this bar; of 62 categories, 19 met it. The remainder are directionally suggestive only and are excluded from the figures below.

Drop in Average Review Score: "Very Late" (7+ days past estimate) vs. "On Time" — by State

State	Score Drop (stars)
PE	-2.66
DF	-2.64
MA	-2.52
RS	-2.51

RJ	-2.51
SC	-2.32
MG	-2.25
SP	-2.21
ES	-2.17
GO	-2.04
CE	-2.03
BA	-2.00
PR	-1.88

Drop in Average Review Score: "Very Late" vs. "On Time" — by Category (top and bottom of the reliable set)

Category	Score Drop (stars)
health_beauty	-2.63
pet_shop	-2.62
perfumery	-2.52
baby	-2.52
sports_leisure	-2.50
furniture_decor	-1.95
stationery	-1.85

Interpretation: across reliable states, the late-delivery penalty ranges from roughly -1.9 to -2.7 stars; across reliable categories, roughly -1.85 to -2.6 stars. The relative narrowness of this range indicates a platform-wide pattern rather than a problem isolated to specific regions or product lines — the fix most likely to move the needle is delivery reliability broadly, not a small number of local interventions.

3.3 Q3 — Seller and Geographic Patterns

Among states with a reliable sample size ($n \geq 50$ delivered, reviewed orders), the ten lowest-scoring were:

State	Mean Score	Orders (n)
MA	3.83	712
AL	3.85	394
SE	3.91	334
PA	3.91	933
BA	3.93	3,229
CE	3.94	1,273
RJ	3.97	12,211
PI	3.99	471
PB	4.08	512
ES	4.08	1,969

The best-performing large states were **SP (4.25, n=40,273)**, **AP (4.24, n=66)**, **AM (4.24, n=144)**, and **PR (4.24, n=4,900)**. Notably, **RJ** — despite being one of Olist's highest-volume states (12,211 orders) — sits among the ten lowest-scoring, which is a meaningful finding on its own: this is not a small, easily-dismissed segment, but one of the platform's largest customer bases underperforming on satisfaction. One state, **RR (n=41)**, did not meet the $n \geq 50$ reliability threshold and is excluded from firm conclusions, though its raw mean (3.90) was directionally similar to the other low-scoring states.

3.4 Q4 — Product Category Performance

Among categories with adequate sample size ($n \geq 10$ delivered, reviewed orders), the ten lowest-scoring were:

Category	Mean Score	Orders (n)
office_furniture	3.65	1,236
fashion_male_clothing	3.82	105
audio	3.85	341
furniture_mattress_and_upholstery	3.89	37
home_comfort_2	3.90	21
home_comfort	3.91	368
fashion_female_clothing	3.92	36
diapers_and_hygiene	3.96	25
fixed_telephony	3.97	209
fashion_underwear_beach	4.01	116

office_furniture stands out as the clearest underperformer — its mean score (3.65) is meaningfully below every other category, and its sample size (1,236 orders) is large enough that this is not a small-sample artifact. The best-performing categories with meaningful volume were **books_general_interest** (4.54, $n=489$) and **food_drink** (4.44, $n=218$); a few of the very top scorers (e.g. **cds_dvds_musicals** at 4.67) have small sample sizes ($n=12$) and should be treated as suggestive rather than conclusive.

3.5 Q5 — Payment Behavior

Average review score by dominant payment type shows a narrow range and no dramatic outlier:

Payment Type	Mean Score	Orders (n)
voucher	4.13	2,953
credit_card	4.16	72,338
boleto	4.16	19,062
debit_card	4.24	1,478

Interpretation: payment type shows only a marginal association with satisfaction (a spread of roughly 0.1 stars across methods) — this is not a meaningful lever for improving review scores on its own, consistent with its low importance in the root-cause model below.

Installments vs. order value: average order value rises steadily with installment count in the well-populated range — from R\$99.94 at 1 installment to R\$183.11 at 6 installments, R\$279.03 at 8, and R\$375.44 at 10 — confirming that customers finance larger purchases over more installments, as expected. Average review score, however, drifts slightly downward over the same range, from 4.19 at 1 installment to roughly 4.04 at 10. Installment counts above 10 involve very small sample sizes (often under 30 orders) and swing erratically (e.g. 24 installments shows 3.33 on only 18 orders), so those extreme tiers should not be read as reliable.

3.6 Q6 — Root Cause Analysis

Collinearity check: `total_order_value` and `total_freight` are moderately correlated with each other ($r = 0.41$), confirming they should be interpreted as one conceptual factor rather than two independent levers. Neither is meaningfully correlated with `delivery_delta` ($r = -0.015$ and -0.050 respectively), so the delivery-timing finding is not an artifact of order size.

Model fit: a Random Forest regression using delivery delay, total order value, total freight, payment installments, payment type, and customer state achieved a held-out test R^2 of **0.117**, meaning the model explains a **modest** share — roughly 12% — of the variation in review scores. This is an important honesty check: these six features are far from a complete explanation of what drives satisfaction on the platform. Product quality, communication, returns

handling, and other unmeasured factors likely account for the majority of the remaining variance, and this model should be read as identifying the strongest available signals, not a comprehensive causal account.

Feature importance: `delivery_delta` was the single largest individual feature at **29.9%** importance, followed by `total_order_value` (27.3%) and `total_freight` (24.1%). Customer state (7.9%) and payment installments (7.5%) trailed well behind, and payment type contributed least (3.3%).

An important nuance on "primary driver": while `delivery_delta` is the single largest individual feature, `total_order_value` and `total_freight` — which are correlated with each other and conceptually represent one factor, "order size" — together account for **51.3%** of importance, larger than `delivery_delta` alone. The most accurate summary is not "delivery is the dominant driver" but rather: *delivery timing and order size are the two strongest associated factors, of roughly comparable overall weight, with delivery timing being the single strongest individual signal and order size being the strongest signal when its two correlated components are considered together.* Geography and payment behavior play a real but distinctly smaller role.

An important measurement caveat: the review survey is triggered once the estimated delivery date has passed, sometimes before the customer has actually received the order. In this dataset, **8.5%** of reviews were submitted before the recorded delivery date. Among that 8.5%, the score distribution is markedly worse — 43.1% were 1-star and only 29.0% were 5-star — compared to reviews submitted after delivery, where only 6.7% were 1-star and 62.0% were 5-star.

Interpretation: this is a meaningful finding in its own right. While only a minority of reviews (8.5%) are affected, that minority is disproportionately negative — customers reviewing before their order arrives appear to be reacting to the wait itself, not the product. This means a portion of the platform's lowest review scores may reflect survey timing rather than final delivery or product experience, and it reinforces (rather than undermines) delivery-related anxiety as a genuine driver of dissatisfaction — but it should be read as a distinct, secondary mechanism alongside actual delivery delay, not folded into it as the same cause.

4. Supporting Visualizations

The following figures, generated in the accompanying Colab notebook, support the findings above:

- Figure 1 — Monthly order volume, revenue, and average review score (Section 3.1)
- Figure 2 — Review score by delivery delay bucket and state/category heatmap (Section 3.2)
- Figure 3 — Bottom 10 states by average review score (Section 3.3)
- Figure 4 — Feature importance chart from the Random Forest model (Section 3.6)

Insert each corresponding chart image (exported via `plt.savefig()` in the notebook) at the marked location above once available.

5. Business Findings

- Delivery timing and order size are the two strongest factors associated with dissatisfaction, of roughly comparable weight — delivery delay is the single strongest individual signal (29.9% importance), while order value and freight combined (51.3%) are the strongest signal when treated as one conceptual factor. Both matter; neither should be presented as the sole driver.
- **MA, AL, SE, PA, BA, CE, RJ, PI, PB, and ES** are the lowest-scoring states with reliable sample size. RJ is a particularly important finding — it is one of Olist's highest-volume states (12,211 orders) yet sits among the ten lowest-scoring, meaning this is not a niche problem.
- **office_furniture** is the clearest underperforming category (3.65 average, $n=1,236$) — its gap below other categories is large and backed by adequate volume, making it a strong candidate for a seller/listing quality audit.

- A meaningful share of low review scores (8.5% of all reviews) are submitted before delivery is confirmed, and this subset skews heavily negative (43% one-star vs. 6.7% one-star for post-delivery reviews) — this is a survey-timing issue that inflates the appearance of dissatisfaction independent of actual delivery outcomes.
- Payment type and installment count show only marginal association with review score and are not meaningful levers for improving satisfaction on their own — the model confirms this with their comparatively low feature importance (3.3% and 7.5% respectively).
- The root-cause model explains only a modest share of review score variation ($R^2 = 0.117$) — this is an honest limitation, not a flaw: it means the six measured features are real, useful signals, but most of what drives satisfaction on this platform is not captured by delivery, order size, geography, or payment behavior alone.

6. Actionable Recommendations

- **Prioritize delivery-estimate accuracy platform-wide.** Since the delay penalty is broad rather than regional (Section 3.2), tightening the accuracy of estimated delivery dates is likely to have wide impact on review scores.
- **Investigate RJ specifically, alongside other high-volume, low-scoring states.** MA, AL, SE, PA, BA, and CE also warrant logistics review, but RJ's combination of high volume and low score makes it the highest-leverage target for improvement.
- **Audit seller and listing quality in office_furniture as a first priority.** Its score gap is the largest in the category data and is backed by adequate sample size; fashion_male_clothing and audio are secondary candidates.
- **Reconsider review survey timing.** Delaying the survey trigger until delivery is confirmed would likely improve data quality — 8.5% of reviews currently reflect pre-delivery anxiety rather than final experience, and that subset is heavily skewed toward 1-star.
- **Treat order size as a genuine, not secondary, satisfaction lever.** Given its combined importance (51.3%) rivals delivery timing, proactive communication and support for higher-value orders deserves the same priority as delivery-reliability initiatives, not a lesser one.
- **Do not over-invest in payment-method or installment-count changes.** Both show weak, marginal association with satisfaction and are unlikely to move review scores meaningfully on their own.

Appendix: Limitations

- Findings are associative, not causal; the data is observational and no experimental manipulation was performed.
- State and category comparisons exclude segments with fewer than the minimum reliable sample size ($n \geq 15$ per delay bucket, $n \geq 10$ for general category/state comparisons, $n \geq 50$ for the state stability ranking); smaller segments may behave differently but cannot be measured reliably with this data.
- The Random Forest model uses a limited feature set (delivery delay, order value, freight, installments, payment type, state) and does not include product-level, seller-level, or review-text signals, which may explain additional variance.
- Review survey timing relative to actual delivery introduces measurement uncertainty into the delivery-satisfaction relationship, as noted in Section 3.6.